COLOR: Cycling, Offline Learning, and Online Representation Framework for Airport and Airplane Detection Using GF-2 Satellite Images

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***Abstract*— Monitoring airports using remote sensing imagery require us to first detect the airports and then perform airplane detection. Detecting airports and airplanes with large-scale remote sensing imagery are significant and challenging tasks in the field of remote sensing. Although many detection algorithms have been developed for detecting airports and airplanes in remote sensing imagery, the efficiency of the processing does not meet the needs of real applications in large-scale remote sensing imagery. In recent years, deep learning techniques, such as deep convolutional neural networks (DCNNs), have achieved great progress in image recognition. However, training a DCNN needs a large number of training examples to accurately fit the data distribution. Annotating training examples in large-scale remote sensing imagery is time-consuming, which makes the pipeline inefficient. In this article, to overcome the above two weaknesses, we propose a novel cycling data-driven framework for efficient and robust airport localization and airplane detection. The proposed method consists of three modules: cycling by example refinement (C), offline learning (OL), and online representation (OR), namely cycling, offline learning, and online representation (COLOR). The OR module is a coarse-to-fine cascaded convolutional neural network, which is used to detect airports and airplanes. The example refinement (ER) module implements the cycling and makes use of the unlabeled remote sensing images and the corresponding predictions obtained by the OR module, to generate training examples. The OL module aims to use the training examples from the ER module to update the OR module, to further improve the performance. The whole workflow involves COLOR. The COLOR framework was used to detect airplanes and airports in 512 large-scale Gaofen-2 (GF-2) remote sensing images with 29 200 × 27 620 pixels. The results showed that the proposed method obtained a mean average precision (mAP) of 88.32% for the airplane detection. In addition due to the proposed coarse-to-fine cascaded OR module the proposed method is much faster than the traditional approaches in real-world applications.**

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***Index Terms*— Data-driven, deep learning, large-scale remote sensing imagery.**

I. Introduction

DEEP learning-based methods always need a large number of annotated images as training samples for remote sensing imagery information extraction. Large-scale remote sensing imagery data sets are very helpful to improve the development of deep learning algorithms for remote sensing imagery information extraction. However, the large-scale annotation is expensive and time-consuming, and it is difficult to annotate pixels for semantic segmentation and objects for object detection in remote sensing images covering wide area, which results in a lack of public large-scale remote sensing imagery data sets.

Some researchers have released annotated remote sensing imagery data sets [1]–[3] to contribute to the development of deep learning based algorithms for remote sensing imagery. However, the trained deep learning models only fit a specific data distribution. These models will perform worse if the images are outside of the fitted data distribution due to the limited generalization ability. Meanwhile, as new remote sensing images are being sampled all the time, a new research problem emerges: how can these trained models fit the new data distribution? Many researchers [4]–[7] have viewed this as a domain adaption problem and have utilized transfer learning techniques to transfer the fitted data distribution to the new distribution. However, these domain adaption algorithms have only been applied to the image classification task. For tasks such as applying domain adaption to airplane detection and airport detection, there has been little related work in remote sensing imagery processing, due to the complexity of these detection tasks, especially for satellite imagery covering a wide area. Many researchers have attempted to apply machine learning algorithms for airplane detection [8]–[13] using small annotated remote sensing image data sets. However, these methods have always been based on handcrafted features. The framework of these methods involves first generating numerous regions of interest (RoIs) using sliding windows or object proposal, and then designing feature descriptors to extract the features, thereby extracting features from the patches. Finally, a classifier is trained to distinguish the background and the airplanes. For example, Bo and Jing [10]

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first performed segmentation to obtain discrete image regions, and then used simple statistical features to detect airplanes; Li *et al.* [11] combined a saliency-based RoI extraction approach with sliding windows to obtain region candidates for airplane detection. Liu and Shi [12] applied radial gradient transform (RGT) descriptors for feature extraction and sparse coding for classification, to detect airplanes. Although the traditional airplane detection methods can obtain relatively impressive results, the feature extraction stage is inefficient. There are two bottlenecks in the feature extraction stage. The first is that designing a feature descriptor is time-consuming, and the other is that extracting features for numerous patches is a slow process. Although the requirement for annotated images has been reduced, the accuracy of these methods is insufficient for real airplane detection with remote sensing imagery covering a wide area due to the poor generalization ability of the handcrafted features. Deep learning is an automatic feature learning and representation framework, which can learn deep features from the data [14]. The convolutional neural network (CNN), as one of the most important methods in image processing, is an efficient and hierarchical feature learning method. Therefore, methods based on CNNs can overcome the above two problems. Many researchers [15]–[18] have now used CNNs to extract features for more efficient and robust airplane detection. However, the performance is dependent on the number of training samples and the consistency of the distribution between the training and test examples. However, remote sensing images acquired at different times will have different data distributions due to the changes in the imaging conditions. Annotating new training samples is thus time-consuming, and the entire pipeline is inefficient.

The imagery used for airplane detection should have a high spatial resolution because the spatial details are important for the detection task. As a result, many types of open-source remote sensing imagery, such as Landsat images, Moderate Resolution Imaging Spectroradiometer (MODIS) images, and so on, cannot be applied in airplane detection. Considering the issues of spatial detail and data availability, the Gaofen-2 (GF-2) satellite panchromatic imagery with a spatial resolution of 1 m can provide fine spatial detail, which is useful for airplane detection. Furthermore, the GF-2 satellite has a wide range of observations and can provide large-scale remote sensing imagery.

However, this large-scale imagery is lacking in annotation for training the algorithm. In order to deal with large-scale remote sensing images, the algorithm should not only be high performance, but also highly computationally efficient and with great generalization ability. Meanwhile, there will be data distribution bias between the images received from the satellite and the data set for study. The algorithm should therefore be dynamically updated, to adapt to the new data distribution. Therefore, detecting airplanes with large-scale remote sensing imagery is a challenging task.

To address these two problems, we propose a novel cycling data-driven framework based on a deep fully CNN (DFCNN) for fast and robust airplane detection with large-scale remote sensing imagery. The framework consists of three modules: cycling by example refinement (C), offline learning (OL), and online representation (OR). The OR module is a coarse-to-fine cascaded CNN, which consists of two sub-networks: an airport classification network (ACN) and an airplane detection network (ADN). The ACN is used to find many coarse airport areas, as fast as possible. The ADN then detects airplanes from the images corresponding to the above candidate airport areas. In general, the network for classification is faster than the one for detection due to the extra layers in the detection network. Therefore, we first apply the ACN to exclude a large number of unrelated areas, and then use the ADN to detect airplanes, for a faster speed and fewer false detections. The cycling is then implemented by the example refinement (ER) module. The ER module is based on active learning, to make use of the unlabeled remote sensing images, and the corresponding predictions as much as possible. Active learning is a special case of semisupervised machine learning and is widely used in medical image analysis [19], and recommender systems [20]. Because the unlabeled data are abundant, and manual labeling is expensive, active learning can help us to more easily acquire training examples. The proposed method is based on active learning to build the ER module for the unlabeled remote sensing images, and the corresponding predictions are produced by the OR module. The OL module aims to use the training samples from the ER module to train the ACN and ADN, to improve the performance. The whole workflow is a cycle between OR, ER, and OL, namely, "cycling, offline learning, and online representation" (COLOR), which breaks the pattern of the single forward workflow in the previous works [8]–[13], [15]–[18]. The COLOR framework is a high-performance and high-efficiency airplane detection framework. The OR module can detect airplanes with high efficiency in large-scale remote sensing images. The ER module continuously adjusts the distribution of the training samples by refining the newly arrived unlabeled remote sensing images. The OL module uses the adjusted training examples to fine-tune the OR module, to more precisely fit the new distribution of the data, thereby boosting the performance.

The main contributions of this article are summarized as follows.

1) To effectively and continually adapt the trained model to new data distribution, the data-driven COLOR framework is proposed. The COLOR framework consists of three modules: C, OL, and OR. The OR module is a task-dependent module and performs the specific recognition task to output the prediction. The cycling is implemented by the ER module, which is responsible for refining the example from coarse to fine with less manual tuning. The fine example is then used in the OL module to tune the models in the OR module. To the best of our knowledge, we are the first to explore a deep learning-based framework from the perspective of the data distribution.
2) Based on the COLOR framework, we propose a cascaded coarse-to-fine method based on DFCNN embedding into the OL module, to rapidly search for airports in remote sensing imagery covering wide area

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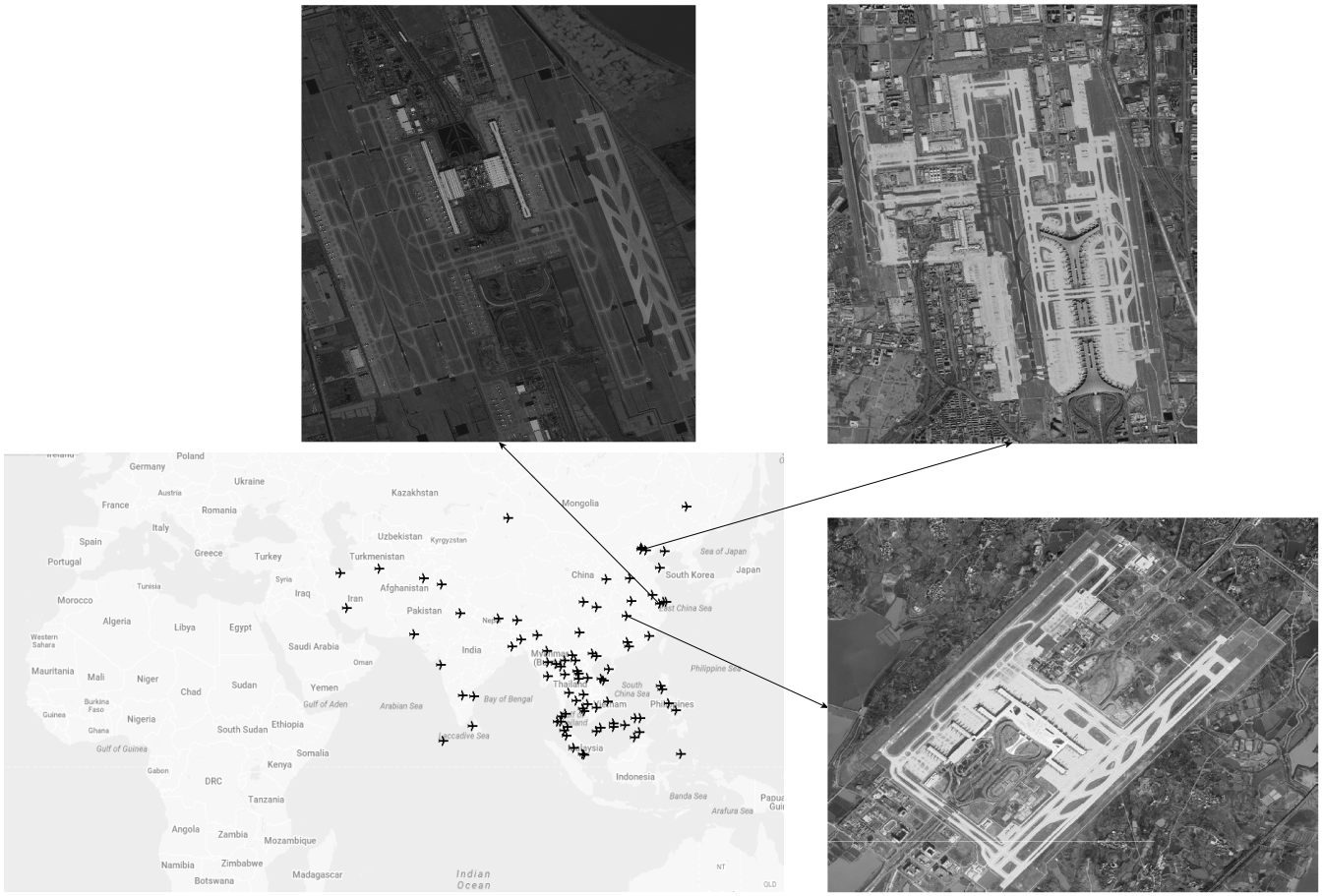


Fig. 1. Distribution of some of the airports all over the world. Black airplane marks: locations of the airports.

and detect airplanes in the searched airport, using only the trained detection model. This can be viewed as a specific implementation of the airplane and airport detection task using the COLOR framework, so we refer to it as Air-COLOR.

3) To validate the effectiveness and efficiency of the COLOR framework, we first collected 512 GF-2 satellite images and applied Air-COLOR to perform automatic airport searching and airplane detection, which obtains a great improvement in speed and accuracy compared with general practice. Benefiting from Air-COLOR, we can obtain high quality bounding box annotations generated by the proposed detection model. In order to promote the development of deep learning-based algorithms for the remote sensing community, we will release the GF-2 data set with the corresponding annotations generated by our model.

II. Data

A. Study Sites and Data Collection

In this article, the areas around airports all over the world were chosen as the study sites for better generalization (Fig. 1). Airplane detection in these areas is valuable for a better analysis of the important hubs. The sparse distribution of the airports in the large-scale remote sensing imagery, and the numerous airplanes in these airport areas, make the automatic detection worthwhile.

The GF-2 satellite is the first civil optical remote sensing satellite independently developed by China. It is equipped with panchromatic (PAN) and multispectral (PMS) sensors with spatial resolutions of 1 and 4 m, respectively. The panchromatic remote sensing imagery was used in our study due to the higher spatial resolution. We chose 512 panchromatic images acquired from the GF-2 satellite with a spatial resolution of 1 m for our study. Each image was of a fixed size of 29 200 $\times$ 27 620 pixels, and the images were collected from 2015 to 2017. The images cover 100 airports and the corresponding surrounding areas all over the world. Therefore, a very wide experimental area is covered by the GF2 data set. Meanwhile, there are a large number of overlapped images and multi-temporal images, so the same airport can appear in different images. For example, Wuhan Tianhe International Airport (WUH) is apparent in many overlapped images and also images collected in both 2015 and 2016.

B. Auxiliary Data

To initialize the parameters of the DFCNN, we used auxiliary data to train the ACN as a starting point. Because the goal of training on auxiliary data is to initialize the OR module for the fundamental discrimination of airports and

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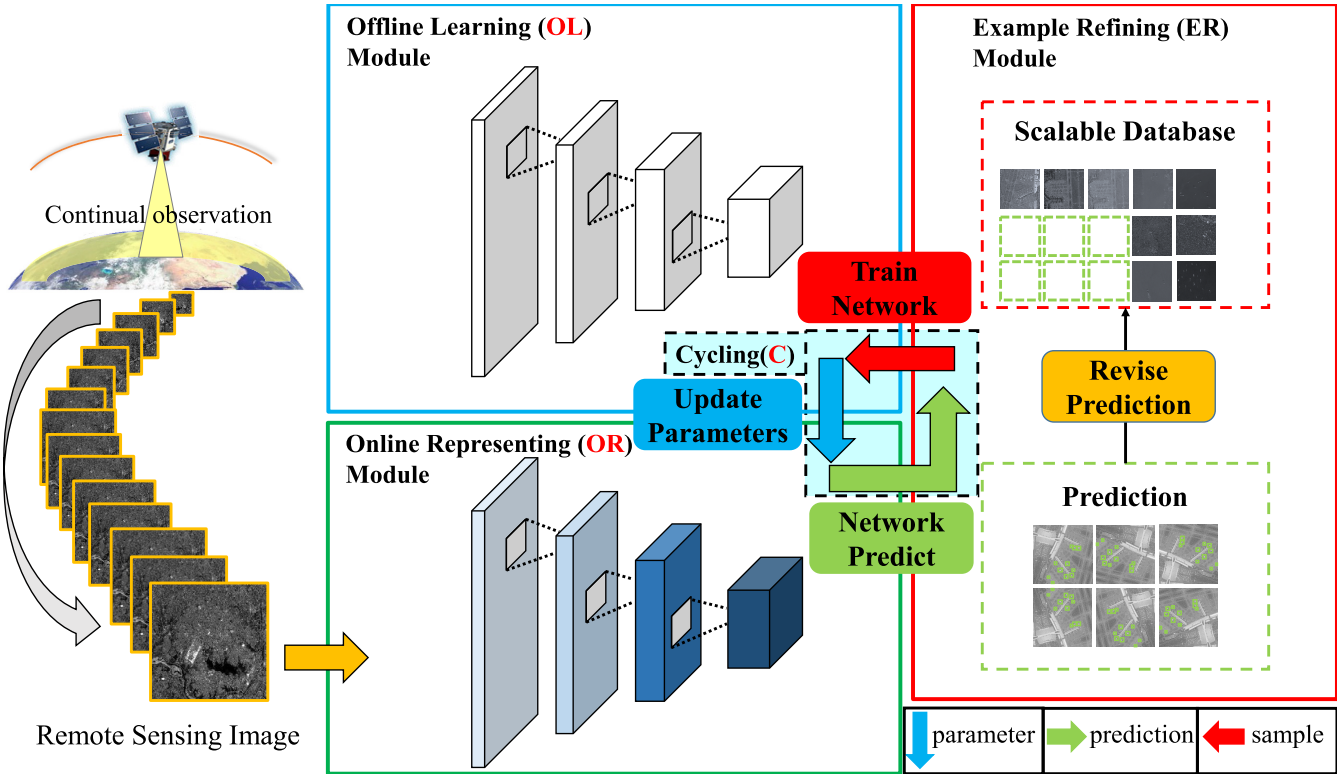


Fig. 2. Overview of the COLOR framework. The OL module passes the parameters of the model to the OR module by training on the ground-truth database. The OR module processes the unlabeled multisource high spatial resolution remote sensing images and obtains predictions that consist of a scene semantic label for the scene classification and many bounding boxes with corresponding semantic labels.

airplanes, the OR module was fine-tuned to more precisely align with the target distribution of the GF-2 data set. The UC Merced (UCM) data set [21] and the NWPU-RESISC45 (NWPU45) data set [22] were chosen to train the ACN to have a better generalization ability, as these two data sets have a large spatial resolution difference. The UCM data set consists of 2100 aerial images of 21 land-use classes, with a pixel resolution of one foot. Each category includes 100 images with a size of 256 × 256 pixels and three channels. The images were manually selected from large images obtained from the USGS National Map Urban Area Imagery collection for various urban areas around the U.S. The NWPU45 data set consists of 31 500 aerial images of 45 scene classes, with the spatial resolution varying from about 30 to 0.2 m/pixel. There are 700 images in each category and each image is in RGB color space and has a size of 256 × 256 pixels with three channels. All the images were collected from Google Earth and labeled by experts [22].

The categories of the UCM data set include airplane, and those of the NWPU45 data set include both airplane and airport. We used the predictions for both airplane and airport as the final predictions for airport because airports obviously have a strong spatial relationship with airplanes.

III. COLOR

An overview of the COLOR framework is given in Fig. 2. To detect airplanes from large-scale remote sensing panchromatic imagery, the DFCNN-based airplane detection framework termed COLOR is proposed. The COLOR framework consists of three modules: C, OL, and OR. The cycling is implemented by the ER module. The pipeline of the COLOR framework contains four steps: 1) training an initial model on manually labeled images; 2) making predictions on unlabeled remote sensing images; 3) selecting some predictions with high confidence to refine and then forwarding these refined samples to the OL module; and 4) retraining the last model according to the fine-tuning on the refined samples. The details of the proposed ER, OL, and OR modules are provided in Sections III-A–III-C.

A. OR Module

The OR module is used to make predictions in the remote sensing panchromatic images from the GF-2 satellite and to provide coarse examples to the ER module, which is deployed on the online environment as a service. It is of great importance to support parallel computing for large-scale remote sensing images. Our COLOR framework is based on DFCNNs, where the parallel tensor computation greatly speeds up the framework. This allows the COLOR framework to handle large-scale remote sensing images efficiently. The OR module, which is a coarse-to-fine cascaded deep CNN (DCNN), contains two subnetworks: the ACN and the ADN. The ACN is responsible for rapid airport localization. The ADN is responsible for accurate airplane detection.

1) ACN: The architecture of the ACN is shown in Fig. 3. The ACN is a variant of ResNet-50 [23], which replaces all the max pooling layers with 3 × 3 convolutional layers with

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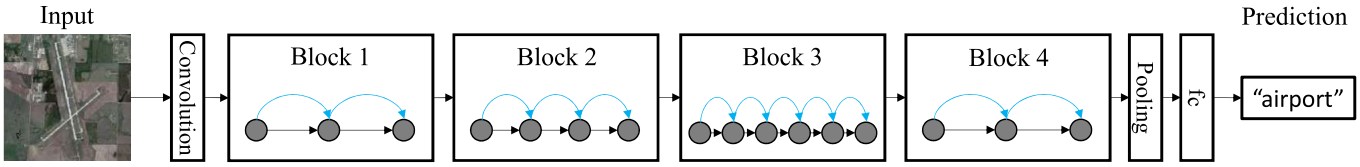


Fig. 3. Network architecture of the ACN.

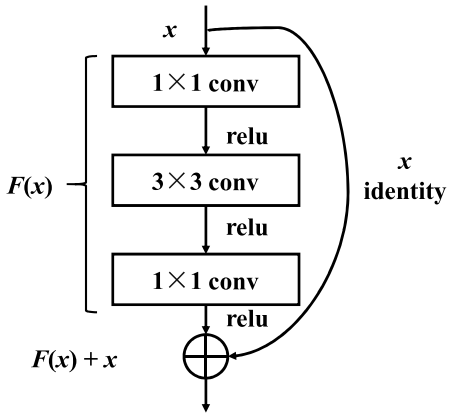


Fig. 4. Module architecture of the bottleneck unit.

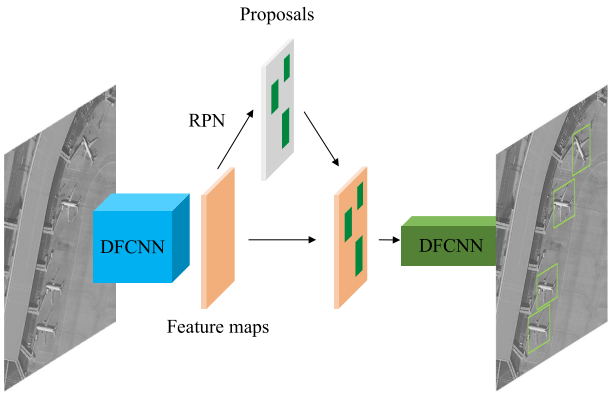


Fig. 5. Architecture of Faster R-CNN.

stride = 2 to keep more spatial information. The ACN follows a modular design, which is made up of four blocks, and these four blocks have three, four, six, and three bottleneck units (Fig. 4), respectively. The convolutional layer is the main component in the bottleneck unit, and consists of an $N \times M$ filter with a $k \times k$ kernel. N denotes the number of channels of the input, and M denotes that of the output. k denotes the size of the kernel. The convolution is a local linear mapping, with the weight shared in the spatial dimension. The bottleneck follows residual learning [23], and learns a residual mapping, rather than direct mapping between the input and output. This makes the optimization of the DFCNN easier and remedies the vanishing gradient problem. The pooling layer and fully connected layer are followed by these four blocks for classification. The pooling layer is a global average pooling layer, which is equal to obtaining the mean value in the spatial dimension mathematically. Compared to the convolutional layer, the fully connected layer is a global linear mapping. In the ACN, the fully connected layer is implemented with a 1×1 convolutional layer to allow images with arbitrary size as the input.

To coarsely locate the airports in a large-scale image with the ACN, we first split the large-scale image into multiple patches with a size of 256×256. The ACN is then applied to classify these patches by batch processing to obtain a coarsely annotated semantic map. Although the annotated semantic map only coarsely reflects the airport areas, it can exclude a large number of irrelevant areas to avoid unnecessary computation. This greatly boosts the efficiency of the whole workflow.

2) ADN: The other network is the ADN, which uses Faster R-CNN [24] as the basic architecture, as Fig. 5 shows. Faster R-CNN is one of the best universal object detection algorithms. It contains two stages: a region proposal stage and a refinement stage. Faster R-CNN first extracts high-level features using the DFCNN, and then applies a region proposal network (RPN) to generate region candidates. In the second stage, namely, the refinement stage, another DFCNN is used to refine these regions for higher localization performance. We use ResNet-50 as the backbone network for Faster R-CNN in the region proposal stage.

3) Coarse-to-Fine DFCNN Cascaded Design: The coarse-to-fine cascaded design is widely used in facial landmark localization [25]–[27] due to the clear correlation between facial landmarks. Considering the semantic relationship between airports and airplanes, we also follow this design to build our OR module, for accurate and rapid airplane detection. Given a large-scale remote sensing image, we first apply the ACN to obtain a coarsely annotated semantic map that indicates the approximate airport areas. We then calculate the minimum bounding rectangle as the proposal and crop the area from the large-scale remote sensing image to obtain a subimage. Finally, the ADN is performed on the subimage to detect airplanes.

B. ER Module

The cycling is implemented by the ER module, which is responsible for revising the coarse examples obtained from the OR module. A coarse example consists of a remote sensing image and the corresponding prediction. The prediction may contain errors, so that we should revise it to be a refined example that can be used to train the model. There are many possible ways to implement the general idea of revision The aim of this article is not to explore a state-of-the-art algorithm for the refining, but simply to show that the workflow in our framework is specially designed for practical applications in

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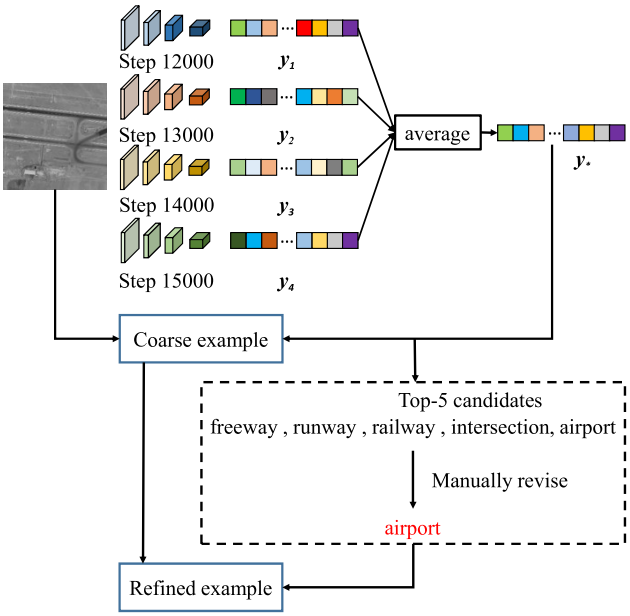


Fig. 6. Example refining for the ACN, where y_i is the predicted probability vector, and y_* is the average of all y_i.

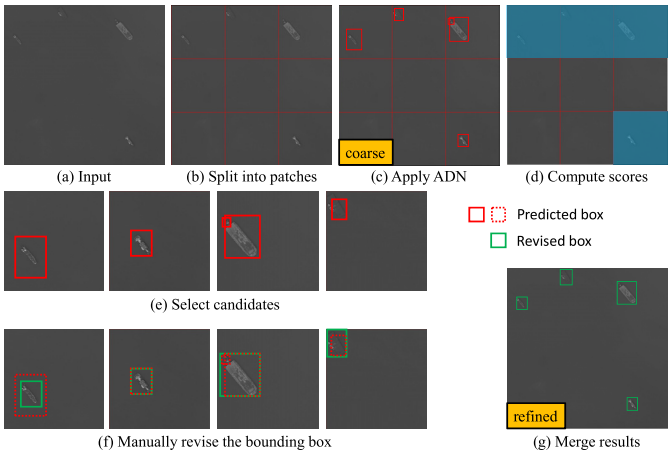


Fig. 7. Example refining for the ADN. (a) Input. (b) Split into patches. (c) Apply ADN. (d) Compute scores. (e) Select candidates. (f) Manually revise the bounding box. (g) Merge results.

remote sensing. Therefore, we adopt a fairly straightforward strategy as follows.

1) The predictions with a high confidence score are kept (score $>$ threshold).
2) The predictions are manually adjusted in the patches by experts.

There are two types of coarse examples. The first consists of the subimage and a probability vector generated by the ACN; the other consists of the subimage and the many bounding boxes with a confidence score and a semantic label generated by the ADN. These two coarse examples are heterogeneous. Therefore, there are two different implementations for the ACN and the ADN, respectively. Based on the above strategy, we propose two different methods to revise the coarse examples for the ACN and ADN, respectively. To provide the ACN with training examples, we first apply the checkpoint ensemble technique [28] to improve the performance of the ACN. Fig. 6 shows the workflow of the ER module for the ACN. For the ACN, the ER module is implemented by top-k class selection and manual revision. The top-k class selection is responsible for reducing the candidate space, which can significantly reduce the cost of the annotation. Essentially, top-k class selection has an adaptive threshold since the threshold is automatically chosen as the predicted probability of the kth class. Compared with using a fixed threshold, this adaptive threshold selection method is more robust. The hyperparameter k controls the number of class candidates, and a larger k will bring a higher recall ratio, but the cost of annotation will increase. For the choice of k, we follow the common practice in the image classification task, where top-1 and top-5 accuracies are generally used as the metrics to judge the performance of the classification model. Therefore, we use top-5 ($k = 5$) class selection, which has enough recall ability to keep most of the true class labels. The refined example can then be obtained by manually selecting the true category from the candidates.

There is a difference in the implementation of the ER module to provide the ADN with training examples, compared to the ACN. The pipeline of the ER module for the ADN is shown in Fig. 7. The ADN is trained using GF2 panchromatic remote sensing images. GF2 panchromatic images have a fixed size of 29 200 $\times$ 27 620 pixels. Due to memory limitations, it is difficult to forward the full image into the ADN. To process the image conveniently, we split the large-scale image into many small patches and then perform the ADN on these patches to obtain the bounding boxes. We then select the many patches with a high confidence score to revise. The confidence score of a patch is computed by the scores of all the bounding boxes. We use the mean of all the scores as the confidence score of the patch, for convenience. The score threshold is set to 0.05, which is common practice [29], [30]. We manually revise these patches to obtain the refined examples. Finally, the separate predictions are merged, and the soft-nonmaximum suppression (Soft-NMS) algorithm [31] is applied as a postprocessing to removing the redundant bounding boxes per class to obtain the complete refined examples.

C. OL Module

The OL is an offline incremental learning module used to fine tune the DFCNN in the OR module using the revised examples from the ER module, which is deployed on the offline environment because it has much slower speed than OR module. The online incremental learning algorithms [32]–[34] are used to train the machine learning models using streaming examples. In our framework, the training examples are from a small part of the manually selected examples and the revised examples provided by the ER module. Therefore, we use a more stable OL algorithm to train the ACN and ADN.

The training set is denoted by

$$D = \{(x_1, y_1), (x_2, y_2), (x_3, y_3), \ldots, (x_N, y_N)\} \quad (1)$$

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TABLE I
CLASSIFICATION RESULTS ON UCM AND NWPU45

Method	Dataset	OA(%)	Airplane OA(%)	Airport OA(%)
ResNet-50 [23]	UCM	91.96	90.00	—
ACN	UCM	92.19	90.79	—
ACN-Fused	UCM	97.32	98.93	—
ResNet-50 [23]	NWPU45	84.47	96.73	82.31
ACN	NWPU45	85.57	96.82	82.78
ACN-Fused	NWPU45	93.80	98.54	84.67

where x_i is the ith image data and y_i is the corresponding target. Because the number of training examples is very large, in practice, we sample a minibatch of training examples to optimize the network. The loss function can be formulated as follows:

$$L(\theta) = \frac{1}{m}\sum_{i=1}^{m} l(x_i, \theta) \tag{2}$$

where θ is the weight of the network, m is the size of the minibatch, and $l(x_i, \theta)$ is the loss computed from the image x_i and corresponding target y_i. Typically, the function l consists of a prediction loss and a regularization loss on θ.

Minibatch stochastic gradient descent (SGD) [35] is applied to update the parameters of the network. The update can be formulated as follows:

$$\theta_{t+1} = \theta_t - \eta\frac{1}{m}\sum_{i=1}^{m} \nabla l(x_i, \theta_t) \tag{3}$$

where η is the learning rate, and t is the iteration index.

The ACN and ADN are separately trained due to their different tasks.

D. Comparison With Previous Frameworks

Here, we discuss the difference between COLOR and the traditional fine-tuning framework. The traditional fine-tuning framework follows a single forward pass pipeline, as shown in Fig. 8(a). This framework needs prepared annotations for new images collected from satellites before fine-tuning. In this way, the new model can adapt to the new data distribution. However, this process is time-consuming and ignores the initial discrimination of the model.

The proposed COLOR framework leverages the initial model to generate coarse examples for the new images in place of the prepared annotations, as shown in Fig. 8(b). The ER module is responsible for revising the coarse examples to fine examples for further offline learning. In this way, the model is continually evolving as the new images flow into the COLOR framework. Meanwhile, apart from obtaining the evolving models, a large-scale data set can be obtained, such as the GF-2 data set, which was generated by the ER module during continual ER.

IV. EXPERIMENTS

To efficiently detect airplanes in the study areas, a coarse-to-fine cascaded method is conducted in the OR module, which uses two subnetworks to first perform airport classification and then detect airplanes, namely, the ACN, and

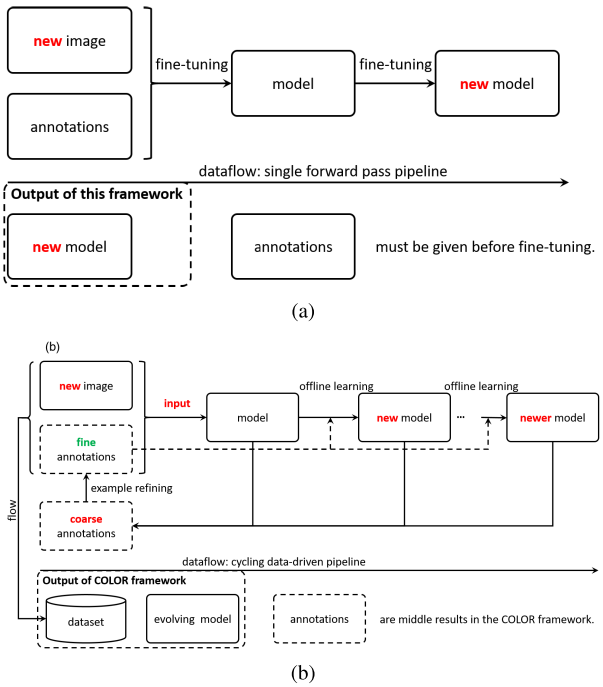


Fig. 8. Comparison between (a) traditional fine-tuning framework and (b) COLOR framework.

TABLE II
NUMBERS OF TRAINING AND TEST EXAMPLES FOR THE DIFFERENT DATA SETS

Dataset	# Train	# Test
UCM	1680	420
NWPU45	6300	25200
GF2-0	1851	—
GF2-1	4911	—
GF2-2	10811	—
GF2-3	—	8685

the ADN. The overall accuracy (OA) is used to evaluate the ACN, as Table I lists, and the mean average precision (mAP) is used to evaluate the ADN. Meanwhile, we conducted many experiments to illustrate the effectiveness and practicality of the ER module. All the experiments were conducted on a server with four NVIDIA Tesla P100 GPU accelerators (with 16-GB GPU memory). All the models were trained from scratch to avoid the influences of different pretrained weights for a fair comparison.

A. Experimental Setup

Table II lists the numbers of training and test examples for the UCM, NWPU45, and GF2 data sets. Note that the GF2 data set has four parts, where GF2-0 is manually labeled and GF2-i ($i = 1, 2, 3$) are generated by the COLOR framework. The complete GF2 data set has 26 258 samples. Because GF2-3 is used as the test data set, extra revision, and inspection was applied to it. Five-fold cross validation was conducted on the experiments with the UCM and NWPU45 data sets.

The ACN and the ADN use Xavier weight initialization [36] for all the convolutional layers. The Xavier weight

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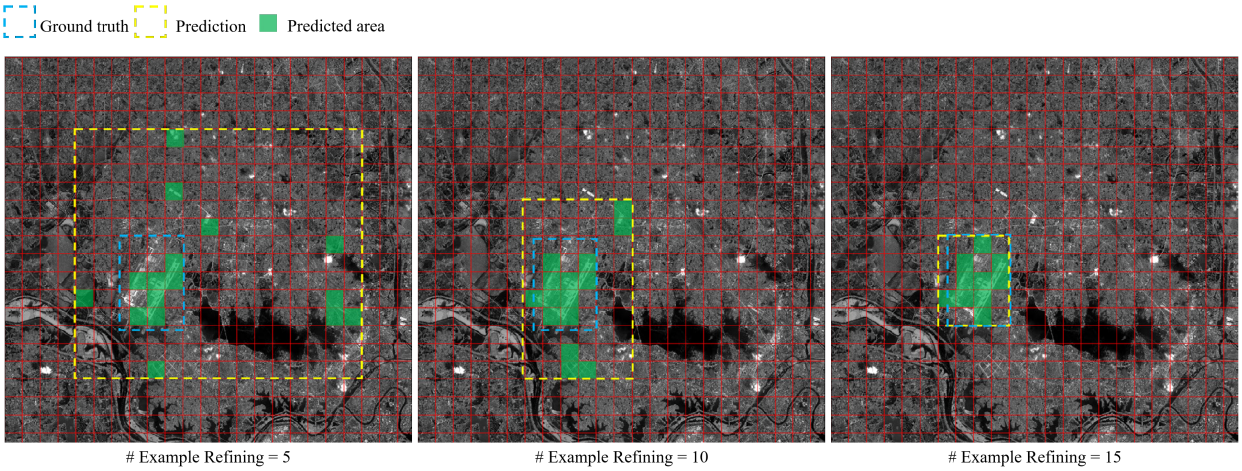


Fig. 9. Performance of airport classification is continually boosted.

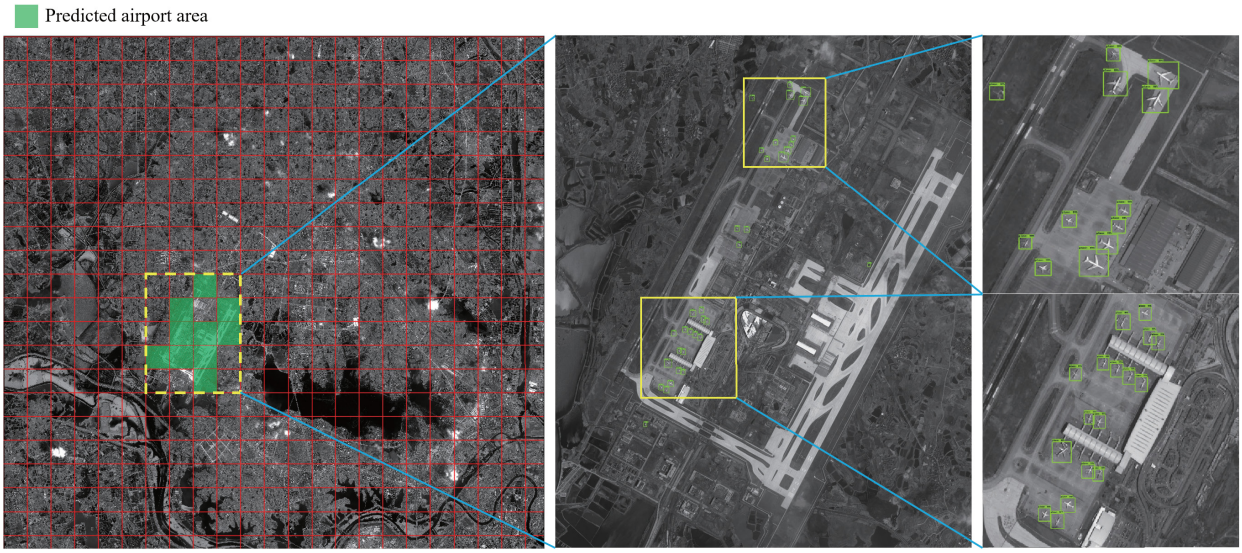


Fig. 10. OR module is applied to search the airport and detect airplanes. The searched airport here is WUH.

initialization makes the variance remain the same for the input and output of each passing layer during the training process, which can keep the signal from exploding to a high value or vanishing to 0. To alleviate the overfitting, there is no bias parameter in the convolutional layers of the backbone of the ADN and the ACN. The fully connected layer in the ACN is implemented by a 1×1 convolutional layer, which follows the above-mentioned weight initialization method.

The ACN was trained using a batch size of 64 for 15 000 steps, and the ADN was trained using a batch size of 16 for 40 epochs. To avoid the introduction of more hyperparameters and make the proposed framework easy to use, SGD optimizer with momentum was used to train the ACN and the ADN. The initial learning rate was set to 0.1.

The momentum was set to 0.9, and the learning rate was divided by 10 at 50% and 80% of the total iteration steps for all the experiments. The weight decay was set to 0.00005.

B. Evaluation of the ER Module in COLOR

We conducted many experiments to validate the effectiveness of the ER module and analyze the effect of the number of rounds of ER on the performance. Fig. 9 suggests that the localization performance is continually improved with more rounds of ER.

We can find that the higher the accuracy of the ACN, the better the localization performance. To improve the accuracy of the ACN, it is important to adjust the distribution of the data by the ER module. The quantitative result of the ACN is shown in Fig. 14. We can see that the performance of the airport classification is continually improving with the increment of the number of rounds of ER. During the early stage of the ER, the gains in accuracy are small but the growth rate gradually increases. Because the coarse examples are gradually changed to fine examples, the parameters of the ACN can be updated by the more accurate gradients computed by the fine examples. During the later stage, the growth rate gradually decreases and the accuracy of the airport classification reaches saturation. Because most of the training examples have completed the ER, the ACN is gradually converged.

Table III suggests that the more rounds of ER, the better the performance. Although GF2-0 is manually labeled, it has a lot of labeling noise. When applying the ER module on GF2-0,

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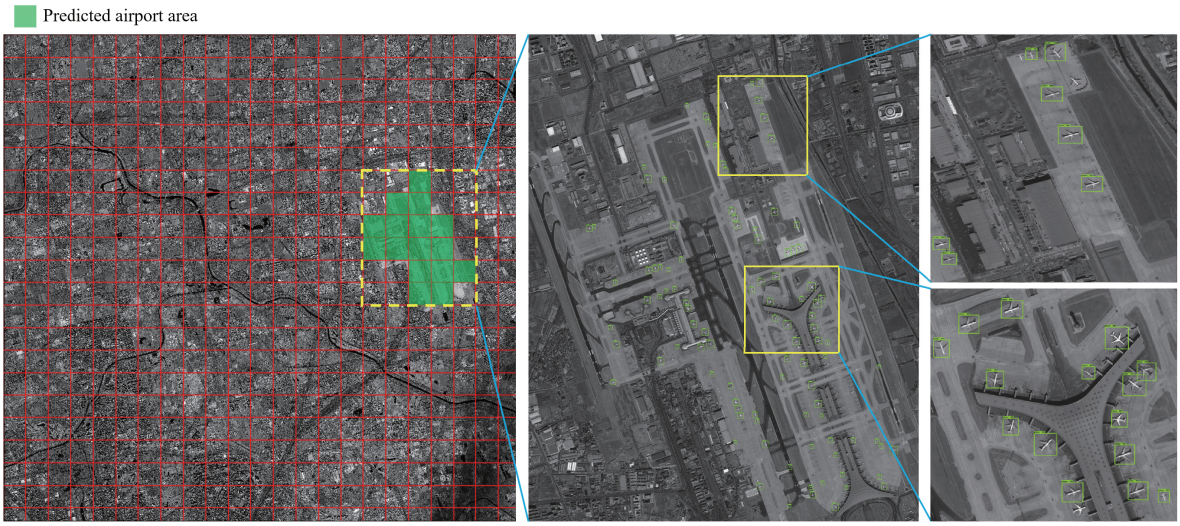


Fig. 11. OR module is applied to search the airport and detect airplanes. The searched airport here is Beijing Capital International Airport (PEK).

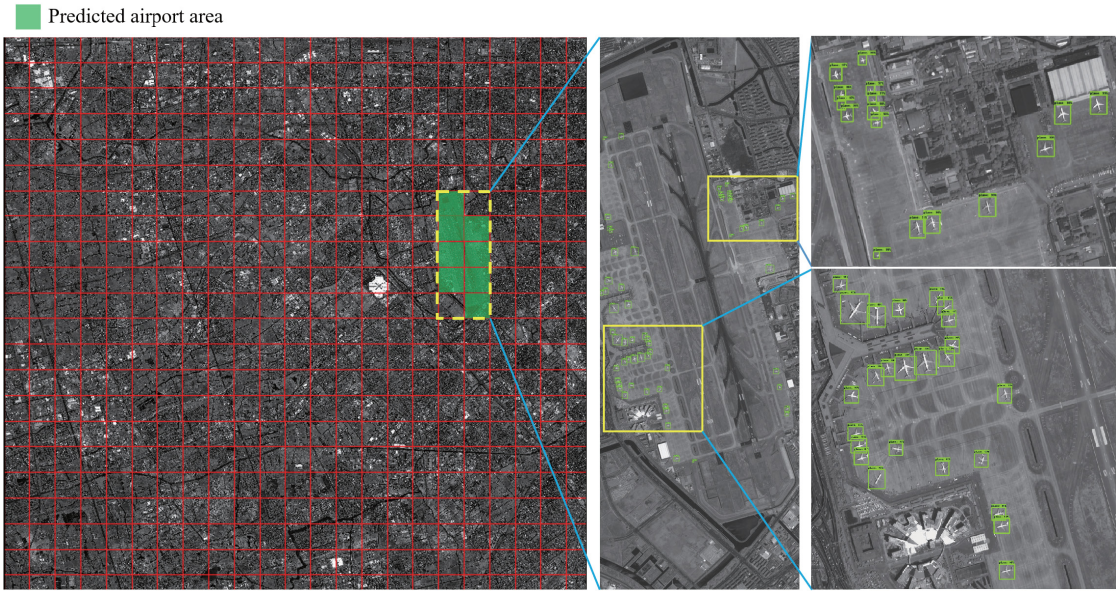


Fig. 12. OR module is applied to search the airport and detect airplanes. The searched airport here is Shanghai Hongqiao International Airport.

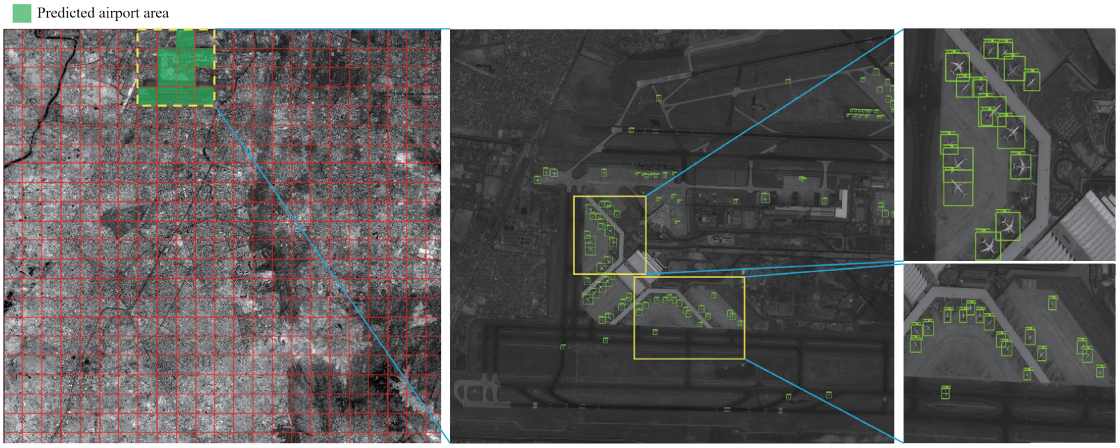


Fig. 13. OR module is applied to search the airport and detect airplanes.

the mAP shows a small improvement from 28.79% to 30.04%. The more important fact is that the detection performance shows a great boost when the ADN is fine-tuned on the revised GF2-0. We also find that introducing new data very much helps to train an ADN with great generalization ability. Fig. 15 shows that the performance of the ADN is continually boosted

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TABLE III

MORE ROUNDS OF EXAMPLE REFINING, THE BETTER THE PERFORMANCE. ER INDICATES EXAMPLE REFINING. # IS THE NUMBER OF ROUNDS OF EXAMPLE REFINING. R-* DENOTES THAT DATA SET * HAS BEEN REFINED

ER on	Training set	#ER	mAP(%)
GF2-0	GF2-0	0	28.79
GF2-0	GF2-0	1	29.48
GF2-0	GF2-0	5	29.74
GF2-0	GF2-0	10	30.04
GF2-0	GF2-0	50	51.72
GF2-1	R-GF2-0	0	54.47
GF2-1	R-GF2-0	1	54.84
GF2-1	R-GF2-0	5	58.63
GF2-1	R-GF2-0	10	61.34
GF2-1	R-GF2-0	50	69.89
GF2-2	R-GF2-0,1	0	73.70
GF2-2	R-GF2-0,1	1	75.64
GF2-2	R-GF2-0,1	5	77.62
GF2-2	R-GF2-0,1	10	78.01
GF2-2	R-GF2-0,1	50	88.32

as the ADN is continually trained using revised examples from the ER module.

C. Evaluation of the OR Module in the COLOR Framework

The classification OA of the ACN is presented in Table I. The ACN keeps more spatial information than ResNet-50, thereby obtaining a performance boost. We used the checkpoint ensemble technique to further enhance the performance, namely ACN-Fused. The results on NWPU45 reflect that recognizing airports is more difficult than recognizing airplanes. Therefore, when applying the ACN to obtain the approximate area of the airports, we merge the prediction of airplanes and airports into the prediction of airports to improve the quality of the airport candidates.

As shown in Table IV, a coarse-to-fine cascaded network (ACN + ADN) is better than directly applying the ADN with sliding windows, both in accuracy and speed. Directly performing the ADN on patches obtained by sliding windows results in a large amount of false positives. However, by using the ACN to filter the numerous redundant areas, the ADN can focus on detecting airplanes in the airport areas, where its background distribution is more consistent with the training set. Meanwhile, the ACN speeds up the progress of the entire framework by greatly reducing the computation time of the ADN. We also compared the proposed method with "you only look once" (YOLO) v3 [37], which is the current state of the art for real-time object detection, to further validate the effectiveness of the proposed method. From (a)/(c) and (b)/(d) in Table IV, we can find that when using YOLO v3 to replace the default implementation of the ADN, the inference speed shows a significant improvement. However, the results indicate that YOLO v3, as a one-stage object detector, lacks the ability to further filter false alarms, so the accuracy is lower than that of the original ADN. We also find that the ACN is able to filter the false alarms, which helps YOLO v3 to achieve higher detection accuracy at a faster speed. This suggests that the coarse-to-fine DFCNN cascaded strategy is efficient and effective. Figs. 10–13 show the results of the airport classification and airplane detection at four airports.

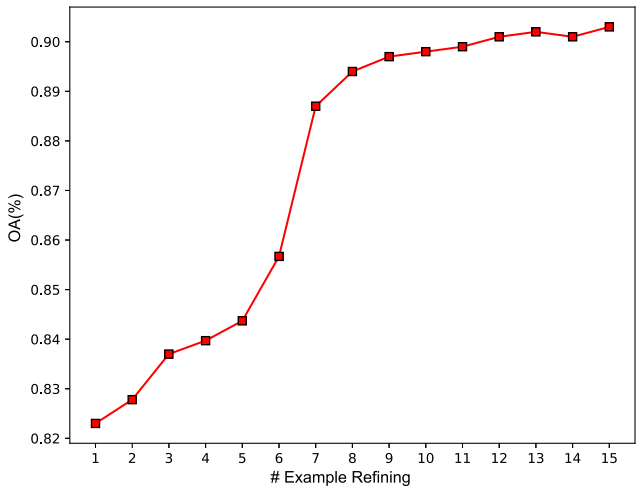


Fig. 14. OA of the ACN during example refining.

TABLE IV

AIRPLANE DETECTION RESULTS ON THE GF2 DATA SET

Method	mAP(%)	Speed(s/scene)
(a) SW + ADN	63.16	72.9
(b) ACN + ADN	88.32	4.3
(c) SW + YOLOv3	52.69	32.24
(d) ACN + YOLOv3	81.65	3.1

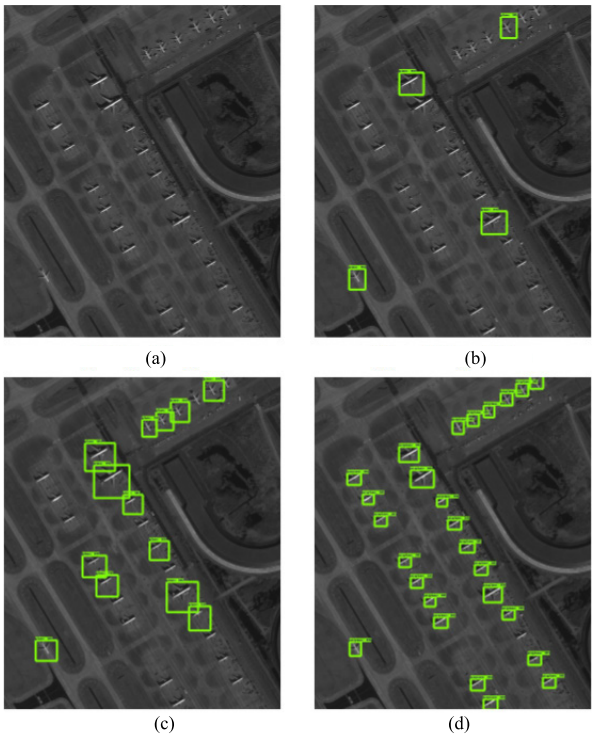


Fig. 15. Model-*i* ($i = 0, 1, 2$) is trained on *GF2-0*, *R-GF2-0* and *R-GF2-0,1*, respectively. (a) Original image. (b) Model-0 prediction. (c) Model-1 prediction. (d) Model-2 prediction.

D. Discussion

The proposed COLOR framework can rapidly locate the airports and detect the airplanes in large-scale high spatial

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resolution remote sensing imagery. Meanwhile, its cycling workflow improves the adaptability for different distributions of data, thereby improving the practicality of the COLOR framework. However, the effectiveness of the airplane detection is highly dependent on the accuracy of the airport localization. Most airports have obvious features, which can be located after several rounds of ER. However, some small airports are difficult to locate, due to the scale problem. This is because learning a scale-invariant feature is difficult using limited examples. For these cases, the OL module can add extra examples with a small scale to remedy the scale problem. If there are a large amount of small airports, the better choice is to use the ER module to adjust the distribution of the data, to improve the performance of the airport localization.

V. Conclusion

In this article, we have presented a novel cycling data-driven framework, namely, COLOR, for airport localization and airplane detection with large-scale GF-2 satellite panchromatic imagery around the world. The COLOR framework consists of three modules: C, OL, and OR. The cycling is implemented by the ER module. The OL module is used to train DFCNNs using the examples from the ER module. The OR module uses the DFCNNs trained by the OL module to perform airport localization and airplane detection. The ER module revises the results of the airport localization and airplane detection to obtain high-quality examples from these coarse examples and provides them to the OL module. The three modules generate a cyclic graph, continually adjusting the training examples and parameters of the DFCNNs, thereby adapting to the various distributions of the imagery data. The COLOR framework overcomes the weakness of the traditional framework (which cannot deal with images with new distributions). We applied the COLOR framework to locate airports all over the world and detect airplanes with large-scale GF-2 satellite panchromatic imagery and analyzed the effectiveness of the ER and OR modules. In our future work, we plan to exploit an effective and robust algorithm for automatic ER to improve the performance in practical applications. Stronger CNN models will also be implemented in the OL module for more robust remote sensing applications. For the generalization of the proposed COLOR framework, we will apply the COLOR framework to more tasks [38]–[40], such as road detection, land-use/land-cover classification, and hyperspectral image processing.

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